

MARSHFIELD, MA, USA

WMO: 722256

Lat: 42.098N

Lon: 70.672W

Elev: 11

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 64774

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.6	(c) 13.9	(d) -8.0	(e) 3.6	(f) 14.1	(g) -3.9	(h) 4.5	(i) 17.6	(j) 25.2	(k) 41.3	(l) 21.3	(m) 39.1	(n) 7.1	(o) 260	(p) 0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	90.1	75.1	86.2	73.1	82.5	71.4	77.2	85.5	75.5	82.7	74.1	80.1	8.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	130.3	81.2	73.1	122.9	78.8	72.3	119.7	78.0	40.6	85.7	39.0	83.4	37.6	80.1	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.4	(b) 17.0	(c) 15.3	(e) 2.2	(f) 94.7	(g) 3.8	(h) 2.8	(i) -0.5	(j) 96.7	(k) -2.7	(l) 98.3	(m) -4.9	(n) 99.8	(o) -7.7	(p) 101.8		
(4) 19.4	(5) 17.0	(6) 15.3	(7) 2.2	(8) 94.7	(9) 3.8	(10) 2.8	(11) -0.5	(12) 96.7	(13) -2.7	(14) 98.3	(15) -4.9	(16) 99.8	(17) -7.7	(18) 101.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	31.1	32.5	38.2	47.2	57.4	65.4	72.6	70.7	64.1	54.3	44.3	36.4
	DBStd	16.38	9.75	8.99	8.21	7.90	8.43	7.42	5.81	5.90	7.29	7.88	8.06	8.99
	HDD50	2318	588	495	382	146	21	0	0	0	45	209	432	
	HDD65	5664	1050	910	833	539	267	85	9	15	104	342	622	888
	CDD50	2786	3	4	15	60	250	462	701	641	424	179	38	9
	CDD65	658	0	0	1	3	32	97	245	191	77	12	0	0
	CDH74	4880	0	0	3	44	291	726	1917	1379	472	48	0	0
	CDH80	1489	0	0	0	12	93	231	639	389	119	6	0	0
	WSAvg	6.7	7.3	7.5	8.2	7.6	6.8	6.2	5.7	5.3	5.9	6.6	7.0	7.0
	PrecAvg	50.8	4.2	3.7	5.4	4.4	4.0	3.7	3.5	4.1	4.1	4.6	4.1	4.7
(15) Precipitation	PrecMax	60.8	7.0	8.0	16.2	9.3	12.3	12.6	8.5	6.5	8.7	12.7	7.4	8.2
	PrecMin	39.2	1.5	2.0	0.4	1.1	1.2	0.1	1.6	0.9	1.0	0.6	0.9	1.5
	PrecStd	6.5	1.5	1.7	3.4	2.2	2.4	3.1	1.7	1.8	2.3	2.9	1.9	2.0
	PrecStd	6.5	1.5	1.7	3.4	2.2	2.4	3.1	1.7	1.8	2.3	2.9	1.9	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.4	63.3	70.2	80.9	88.5	91.5	93.1	92.6	89.9	80.7	69.8	63.3
		MCWB	58.6	56.8	55.4	63.9	68.2	77.0	76.8	78.0	76.1	69.0	64.2	60.3
	2%	DB	55.4	55.2	61.4	71.7	81.7	86.1	90.2	87.9	82.4	73.2	64.3	59.1
		MCWB	51.4	50.2	52.3	59.9	67.1	72.1	75.5	74.5	72.2	66.9	59.3	55.6
	5%	DB	51.6	51.6	55.2	64.3	75.3	81.6	86.5	84.0	79.4	70.2	61.3	54.8
		MCWB	48.2	47.3	49.0	55.6	65.1	69.8	73.5	72.7	70.2	64.9	55.2	51.4
	10%	DB	45.5	45.5	50.3	60.7	70.5	77.4	83.6	81.4	75.2	66.1	57.0	50.3
		MCWB	41.5	40.8	44.6	52.4	61.9	66.6	72.3	71.6	69.2	61.6	52.0	47.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.0	58.2	59.5	67.0	72.7	77.4	79.4	79.0	77.2	72.5	65.8	62.3
		MCDB	61.3	61.8	64.3	72.3	83.9	89.1	90.7	89.7	84.9	75.9	68.5	63.0
	2%	WB	53.9	52.3	54.5	62.2	69.7	74.4	77.4	76.9	74.7	68.9	61.8	56.9
		MCDB	55.7	54.7	59.1	69.7	78.7	82.7	86.5	83.7	80.6	72.1	63.9	58.7
	5%	WB	47.2	46.4	50.4	58.0	66.9	72.4	75.6	75.3	72.8	65.7	57.0	52.3
		MCDB	49.2	49.6	56.1	63.9	73.7	78.8	83.2	81.6	77.5	69.7	60.7	54.4
	10%	WB	42.1	41.3	45.1	53.6	63.6	70.1	74.2	73.5	70.5	62.5	53.6	47.3
		MCDB	45.4	45.0	48.5	58.0	68.0	75.3	80.3	78.7	74.7	65.0	56.0	49.3
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	15.2	13.8	16.3	16.3	16.0	16.4	16.6	16.3	16.1	15.1	13.9
		MCDBR	19.9	21.1	22.1	24.1	24.4	21.1	20.8	19.5	19.5	18.7	19.2	18.5
		MCWBR	18.9	16.6	15.3	14.7	12.6	11.3	9.4	9.2	11.8	12.9	15.3	17.7
	5% WB	MCDBR	19.2	20.4	20.7	21.4	22.4	19.2	17.6	16.9	16.8	17.3	17.0	17.9
		MCWBR	19.2	18.0	16.4	15.8	12.8	12.1	9.3	8.9	11.4	13.8	16.0	18.6
(40) Clear-Sky Solar Irradiance	taub		0.300	0.308	0.329	0.371	0.407	0.458	0.479	0.437	0.377	0.340	0.315	0.301
	taud		2.515	2.503	2.489	2.395	2.337	2.231	2.178	2.288	2.437	2.540	2.571	2.568
	Ebn at Noon		271	287	292	285	276	261	254	262	272	271	264	262
	Edh at Noon		23	27	30	36	39	44	46	40	32	26	22	20
(44) All-Sky Solar Radiation	RadAvg		553	832	1154	1492	1674	1804	1885	1695	1365	917	616	462
	RadStd		40	76	83	160	173	167	111	76	92	78	45	50

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+79

Nomenclature: See separate page